

The PINO loss as a least-squares problem: structure, conditioning, preconditioning

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Working notes, not paper material. The question: PINO’s `mse` reduction looks like it should be *formally* the same object as our bare least-squares arm, differing only in which discrete Laplacian appears. This confirms that, identifies the matrix precisely, works out its spectrum and conditioning, asks how one would precondition it, and then treats the `rel` reduction that PINO actually ships — whose geometry is not least-squares at all.

Every number below is computed at the paper’s resolution, $n_x = n_y = 65$ on the unit square ($h = 1/64$, $N = 63^2 = 3969$ interior nodes), with the repo’s own assembly (`PoissonProblem`) for the finite-element operator. Analytic and numerical spectra agree to six digits, so the closed forms quoted here can be trusted.

1 Setup

The PDE is $-\Delta u = f$ on $\Omega = (0,1)^2$ with $u|_{\partial\Omega} = 0$. Write $u \in \mathbb{R}^N$ for the vector of interior nodal values and $f \in \mathbb{R}^N$ for the source. Both arms impose the boundary condition *hard, in the model*: our arms through `apply_zero_boundary`, PINO (since the switch away from the mollifier) through `ZeroBoundaryModel`, which is the same operation. So in both cases the boundary values are exactly zero and can be eliminated, leaving a square system on the interior.

Our least-squares arm. With A_{FE} the Q_1 stiffness matrix and M the consistent mass matrix,

$$L_{\text{LS}}(u) = \frac{1}{2} \|A_{\text{FE}}u - Mf\|_2^2. \quad (1)$$

PINO, mse reduction. The Laplacian is taken with second-order central differences and the residual is averaged over interior points only:

$$L_{\text{mse}}(u) = \frac{1}{N} \sum_{i,j \text{ interior}} (-\Delta_h u - f)_{ij}^2 = \frac{1}{N} \|A_{\text{FD}}u - f\|_2^2, \quad (2)$$

where A_{FD} is the five-point Laplacian $(A_{\text{FD}}u)_{ij} = h^{-2} (4u_{ij} - u_{i\pm 1,j} - u_{i,j\pm 1})$.

2 The mse variant is a least-squares problem

Comparing (1) and (2): **yes, formally identical in structure** — both are $\frac{c}{2} \|Au - b\|_2^2$ for a symmetric positive-definite A and a fixed right-hand side, on a single sample. The differences are entirely in *which* (A, b, c) .

That statement can be sharpened, because the five-point stencil is itself a finite-element operator. Scaling out the h^{-2} ,

$$h^2 A_{\text{FD}} = \begin{bmatrix} & -1 & \\ -1 & 4 & -1 \\ & -1 & \end{bmatrix} = A_{P_1}, \quad (3)$$

the stiffness matrix of *piecewise-linear P_1 elements on the uniform right-triangulation* of the same grid (verified numerically: $h^2 A_{\text{FD}}$ has entries 4 and -1 exactly). Likewise, f sampled pointwise is h^{-2} times the *lumped* load $M_L f$ with $M_L = h^2 I$ — the consistent mass matrix M has interior row sums exactly h^2 , so lumping replaces Mf by $h^2 f$. Therefore

$$L_{\text{mse}}(u) = \frac{h^{-4}}{N} \|A_{P_1} u - M_L f\|_2^2 \quad (4)$$

and PINO’s `mse` loss is exactly our least-squares loss with three substitutions:

1. Q_1 bilinear quadrilaterals $\rightarrow P_1$ linear triangles (nine-point $\rightarrow$ five-point);
2. the consistent mass matrix $\rightarrow$ a lumped one (equivalently: exact quadrature of the load $\rightarrow$ pointwise sampling);
3. an overall constant $\frac{1}{2} \rightarrow h^{-4}/N$.

At $n = 65$ that constant differs by a factor $2h^{-4}/N \approx 8.5 \times 10^3$. Under plain gradient descent this would demand a correspondingly smaller step; under Adam it very largely cancels, since Adam divides by a running RMS of the gradient. This is worth remembering when reading the learning-rate sweeps: the reductions differ by orders of magnitude in scale and the sweeps still land within a factor of a few of one another, precisely because the optimizer is scale-free.

The third substitution is bookkeeping. The first two are real, and they are what the rest of this note is about.

3 Properties of A_{FD}

3.1 Structure

A_{FD} is symmetric positive definite, block-tridiagonal with tridiagonal blocks, and has the Kronecker form $A_{\text{FD}} = h^{-2}(T \otimes I + I \otimes T)$ with $T = \text{tridiag}(-1, 2, -1)$. It is block-Toeplitz-Toeplitz-block: the stencil is constant across the grid, which is a consequence of the uniform mesh and would be lost on a graded or unstructured one. Bandwidth is $n_x - 2$; five nonzeros per interior row against nine for A_{FE} .

3.2 Spectrum: both operators share an eigenbasis

On a uniform grid with homogeneous Dirichlet conditions the discrete sine functions

$$v_{ij}^{(k,l)} = \sin(k\pi ih) \sin(l\pi jh), \quad k, l = 1, \dots, n_x - 2, \quad (5)$$

are *exact* eigenvectors of both operators. This is not an approximation and not special to finite differences: it follows from the constant stencil plus the odd reflection at the boundary. Verified numerically — applying either matrix and comparing against the diagonal action in DST-I space gives a relative error of 5.9×10^{-16} (A_{FE}) and 3.4×10^{-16} (A_{FD}). Consequently both are diagonalised

by the same fast sine transform, and every spectral statement below is available in closed form. With $\theta_k = k\pi h$:

$$\lambda_{kl}^{\text{FD}} = \frac{4}{h^2} \left(\sin^2 \frac{\theta_k}{2} + \sin^2 \frac{\theta_l}{2} \right), \quad (6)$$

$$\lambda_{kl}^{\text{FE}} = \frac{1}{3} \left(8 - 2 \cos \theta_k - 2 \cos \theta_l - 4 \cos \theta_k \cos \theta_l \right). \quad (7)$$

3.3 Conditioning

	$\lambda_{\min}$	$\lambda_{\max}$	$\kappa(A)$
$A_{\text{FE}} (Q_1, 9\text{-point})$	4.816×10^{-3}	3.997	829.9
$A_{\text{FD}} (5\text{-point})$	19.74	3.275×10^4	1659.4

Both scale as $\kappa(A) = \Theta(h^{-2})$, as any second-order elliptic discretisation must. The interesting part is the constant: at this resolution

$$\frac{\kappa(A_{\text{FD}})}{\kappa(A_{\text{FE}})} = 1.9996 \approx 2, \quad (8)$$

i.e. **the five-point Laplacian is twice as ill-conditioned as the Q_1 stiffness matrix**. The reason is visible in the symbols: at the highest frequency $\theta = \pi$ the FD symbol reaches $8h^{-2}$ while the Q_1 symbol saturates at 4 (in its own scaling) — the nine-point stencil’s corner couplings flatten the top of the spectrum, while the bottom of the spectrum is fixed for both by consistency with $-\Delta$. Note also the enormous difference in *absolute* scale: $A_{\text{FD}} = \Theta(h^{-2})$ against $A_{\text{FE}} = \Theta(1)$, which is the h^{-4} in (4) showing up again.

For optimisation what matters is the Hessian of the loss. For a linear model $\nabla^2 L_{\text{LS}} = A^\top A = A^2$, so

$$\kappa(\nabla^2 L) = \kappa(A)^2 = \begin{cases} 6.89 \times 10^5 & (Q_1 \text{ weak form}) \\ 2.75 \times 10^6 & (\text{five-point FD}) \end{cases} \quad (9)$$

a factor 4 apart. The square is the same mechanism that motivates preconditioning in the paper; the strong form simply makes it worse.

4 How one would precondition A_{FD}

Everything we do for A_{FE} transfers, because A_{FD} is the more classical object of the two — textbook multigrid was developed for exactly this operator.

Geometric multigrid. A V-cycle with damped-Jacobi smoothing, bilinear prolongation and $R = P^\top$ applies unchanged. Only the smoother parameter differs. The optimal damping is $\omega_{\text{opt}} = 2/(\mu_{\min}^{\text{HF}} + \mu_{\max})$ where μ are the eigenvalues of $D^{-1}A$ and HF denotes the modes not representable on the coarse grid. Computed:

	ω_{opt}	smoothing factor μ
$A_{\text{FE}} (Q_1, 9\text{-point})$	$0.8892 \approx 8/9$	0.333
$A_{\text{FD}} (5\text{-point})$	$0.8002 \approx 4/5$	0.599

Both reproduce the textbook values, which is a useful check on the method we used to derive $8/9$ for Q_1 . The second column is the more interesting one: **Jacobi is a markedly worse smoother for the five-point stencil** — one sweep removes a factor 0.6 of the high-frequency error against $1/3$ for Q_1 . Recovering the same per-cycle error reduction therefore costs roughly twice as many smoothing steps, since $0.6^{2.2} \approx 0.13$ against $(1/3)^{2.2} \approx 0.012$. The nine-point stencil’s corner couplings, which cost more arithmetic per matrix-vector product, buy a genuinely better smoother.

Exact spectral preconditioning. Since the eigenvectors are known and the transform is fast, A_{FD}^{-1} can be applied *exactly* in $O(N \log N)$ by DST: transform, divide by λ_{kl}^{FD} , transform back. On a uniform square this is strictly better than multigrid — it is a direct solver at near-optimal cost, and $\kappa(PA_{\text{FD}}) = 1$ by construction. The reason we do not use it, and the reason it should not tempt anyone, is that it exists only for constant-coefficient problems on tensor-product domains with these boundary conditions. Multigrid survives variable coefficients, unstructured meshes and general geometry; the DST does not. If a spectral preconditioner were used in the paper it would be measuring the wrong thing, in the same way the $\sin(\pi x) \sin(\pi y)$ mollifier was.

Cheaper options. Incomplete Cholesky, or a symmetric Gauss–Seidel sweep, both apply and both leave the $\Theta(h^{-2})$ growth intact — they improve the constant, not the rate. Only a multilevel method (or the exact transform) gives h -independent conditioning.

5 Finite differences against the weak form

	A_{FD} (strong form)	A_{FE} (weak form)
stencil	5-point	9-point (Q_1)
scaling	$\Theta(h^{-2})$	$\Theta(1)$
κ at $n = 65$	1659	830
smoothing factor	0.60	0.33
right-hand side	f pointwise	Mf , exact quadrature
eigenvectors	discrete sines — shared	
symmetry / definiteness	SPD — shared	
consistency order	$O(h^2)$ — shared	

Beyond the table, three differences are conceptual rather than numerical.

Variational characterisation. $A_{\text{FE}}u = Mf$ is the Galerkin condition: u_h is the H_0^1 -orthogonal projection of the true solution onto the finite-element space, and the residual $A_{\text{FE}}u - Mf$ is the gradient of the energy $E(u) = \frac{1}{2}u^\top A_{\text{FE}}u - u^\top Mf$. This is what makes the Deep Ritz arm possible at all, and it is what gives the natural B -norm in which the preconditioned residual should be measured. The finite-difference residual has no such characterisation — A_{FD} is a consistent approximation of the operator, not the representation of a bilinear form in a basis, and there is no energy whose gradient it is. (One may of course *reverse-engineer* one via the P_1 identification (4), which is exactly what that identity says; but the strong form as written does not supply it.)

The right-hand side. Mf integrates the source against the basis functions; pointwise f samples it. For smooth f both are second-order accurate and the difference is a lower-order perturbation. For rough or high-frequency f they are not equivalent, and pointwise sampling is the more fragile

of the two — it can alias. With $K = 10$ modes on a 65^2 grid we are far from that regime, so this difference should not be visible in our experiments; it would be at K comparable to $n_x/2$.

Boundary treatment. Because we now zero boundary nodes in both arms, the Dirichlet elimination is identical and this is no longer a difference. Under the mollifier it was: the mollified model could not represent any function that failed to vanish like $\sin(\pi x)\sin(\pi y)$ near the boundary, which is a constraint on the hypothesis class, not on the discretisation.

6 The rel reduction: not a least-squares problem

What PINO actually ships, and what we are running, is

$$L_{\text{rel}}(u) = \frac{\|A_{\text{FD}}u - f\|_2}{\|f\|_2} \quad (\text{averaged over the batch}), \quad (10)$$

a *norm*, not a squared norm. This single square root changes the geometry completely, and the change is not a matter of scaling.

6.1 The loss surface is a cone

Write $r = A_{\text{FD}}u - f$. Then $L_{\text{rel}} = \|r\|/\|f\|$ is positively homogeneous of degree one in r : along any ray towards the minimiser the loss decreases *linearly*, not quadratically. Near $r = 0$ the surface is a cone with a point at the bottom, not a paraboloid, and L is not differentiable at the minimum.

The gradient in residual space is

$$\frac{\partial L_{\text{rel}}}{\partial r} = \frac{r}{\|r\| \|f\|} \implies \left\| \frac{\partial L_{\text{rel}}}{\partial r} \right\| = \frac{1}{\|f\|} \quad \text{exactly, for every } r \neq 0. \quad (11)$$

Measured across six decades of $\|r\|$ (from 6.3×10^1 down to 6.2×10^{-5}) the gradient norm is constant at $1.5799 \times 10^{-2} = 1/\|f\|$ to all printed digits. In parameter space,

$$\frac{\partial L_{\text{rel}}}{\partial u} = \frac{A_{\text{FD}} r}{\|r\| \|f\|}, \quad \left\| \frac{\partial L_{\text{rel}}}{\partial u} \right\| = \frac{\|A_{\text{FD}} \hat{r}\|}{\|f\|} \in \left[\frac{\lambda_{\min}}{\|f\|}, \frac{\lambda_{\max}}{\|f\|} \right], \quad (12)$$

so it is bounded away from zero but direction-dependent — it does not vanish at the solution either.

Consequence. A method with a fixed step size *cannot converge* on this loss. The gradient does not shrink as the residual does, so the iterate overshoots by a fixed distance and settles into a limit cycle whose radius is proportional to the step size — the same phenomenon measured earlier for Adam on the quadratic loss ($r_{\infty} \approx 0.011 \text{ lr}$), but here it is forced by the loss rather than by the optimizer’s variance estimate. Convergence *requires* annealing the learning rate, which the cosine schedule supplies. Anyone running this loss at constant learning rate should expect a floor and should not read it as the method’s accuracy.

6.2 Conditioning

The Hessian in residual space is

$$\frac{\partial^2 L_{\text{rel}}}{\partial r^2} = \frac{1}{\|r\| \|f\|} (I - \hat{r} \hat{r}^{\top}), \quad \hat{r} = r/\|r\|, \quad (13)$$

and pulling back to u ,

$$\nabla_u^2 L_{\text{rel}} = \frac{1}{\|r\| \|f\|} A_{\text{FD}} (I - \hat{r} \hat{r}^\top) A_{\text{FD}}. \quad (14)$$

Three things follow, and they matter more than the $\kappa(A_{\text{FD}})^2$ of the `mse` case.

1. **It is singular.** The projector kills the radial direction, so $\nabla^2 L_{\text{rel}}$ always has a null direction. The condition number, taken literally, is infinite at every point. The meaningful quantity is the conditioning *on the tangent space*.
2. **On that tangent space the conditioning is unchanged.** Restricted to $\hat{r}^\perp$, (14) is A_{FD}^2 up to the scalar $1/(\|r\| \|f\|)$, so the spread of curvatures is still $\kappa(A_{\text{FD}})^2 \approx 2.75 \times 10^6$. **Normalising by $\|f\|$ does not precondition anything.** It rescales the loss and flattens one direction; the ill-conditioning inherited from the discrete Laplacian is untouched. Preconditioning the `rel` loss still requires a $P \approx A_{\text{FD}}^{-1}$ inside the norm, exactly as for the squared loss.
3. **The scale diverges as training proceeds.** The prefactor $1/\|r\|$ grows without bound as $r \rightarrow 0$. So the loss becomes *stiffer* the closer one gets, while the gradient norm stays fixed — the ratio of curvature to gradient blows up like $1/\|r\|$. This is the precise sense in which the `rel` loss behaves like normalised gradient descent, and it explains the empirical observation that the mollified PINO arm was nearly insensitive to the learning rate across three decades (0.32 – 0.48 best validation error for $\text{lr} \in [10^{-4}, 3 \times 10^{-2}]$): the loss self-normalises the gradient scale, so the learning rate mostly sets the limit-cycle radius rather than the rate of progress.

6.3 The per-sample normalisation

Dividing by $\|f\|$ *per sample* before averaging over the batch equalises the contribution of samples with different source amplitudes. Our summed convention does not: a sample with a large $\|f\|$ dominates the batch gradient. Which is preferable is a modelling choice rather than a mathematical fact — but it is a third axis on which the two objectives differ, independent of strong-versus-weak form and of finite differences versus assembly, and it should not be conflated with either.

7 Summary

- PINO’s `mse` reduction *is* a least-squares loss, and specifically (4): our least-squares loss with P_1 triangles instead of Q_1 quads, a lumped instead of a consistent mass matrix, and a constant h^{-4}/N in front.
- Its matrix A_{FD} shares an eigenbasis with A_{FE} (discrete sines, exactly), is SPD, and is $\Theta(h^{-2})$ -conditioned like A_{FE} — but with twice the constant, hence four times the Hessian condition number.
- It is preconditioned the same way. Geometric multigrid applies verbatim with $\omega_{\text{opt}} = 4/5$ instead of $8/9$; damped Jacobi is a distinctly worse smoother on the five-point stencil (0.60 against 0.33).
- The `rel` reduction that PINO actually uses is *not* least squares. It is conically non-smooth at the minimum, its gradient norm is exactly $1/\|f\|$ everywhere, it cannot converge at a fixed step size, and — the point most likely to be misread — **its normalisation does not improve conditioning**. On the tangent space the curvature spread is still $\kappa(A_{\text{FD}})^2$.